

PHASE 2 – REQUIREMENT ANALYSIS

1.INTRODUCTION

The Requirement Analysis phase identifies the functional and technical requirements of the module "Import Data using Transform Maps (Spreadsheets)." The main purpose is to define how ServiceNow should import student and faculty records into the Institution Details table using staging tables and transform maps.

2.FUNCTIONAL REQUIREMENTS

2.1 Data Import Requirement

A spreadsheet (Excel/CSV) shall be prepared containing sample records with the following fields:

- Student Roll Number
- Student Name
- Faculty Name
- Branch (ECE, EEE, CSE)
- Email
- Phone Number
- Description

2.2 Transform Map Requirements

A custom staging table and transform map shall be created:

Entity	Name	Purpose
Staging Table	u_institution_details_import	Holds raw uploaded data from the spreadsheet.
Transform Map	Institution Details Transform Map	Maps staging fields to the target u_institution_details table.

2.2.1 TABLE & FIELD MAPPING REQUIREMENTS

The mapping from the source spreadsheet/staging table to the target table shall be configured as follows:

Source Field (Import Set)	Target Field	Target Field Type
u_student_roll_number	u_student_roll_number	Auto Number
u_student_name	u_student_name	Reference - User
u_faculty_name	u_faculty_name	Reference - User
u_branch	u_branch	Choice (ECE, EEE, CSE)
u_email	u_email	String
u_phone_number	u_phone_number	String
u_description	u_description	Multi-line String

3. DATA MAPPING & VALIDATION REQUIREMENTS

Field Name	Handling Rule
Reference Fields	Student Name and Faculty Name must map to existing sys_id or user_name entries in the User (sys_user) table.
Choice Fields	Branch values must exactly match existing choices (ECE, EEE, CSE).
Coalesce Rule	u_email or u_student_roll_number may be set as Coalesce = true to prevent duplicate record insertion.

4.SECURITY REQUIREMENTS

The system shall:

- Require administrator or admin / import_transformer roles to execute data loads and run transform maps.
- Respect existing ACL restrictions on u_institution_details after data is transformed into target records.
- Prevent unauthenticated or unauthorized users from loading files into staging tables.

6. SOFTWARE REQUIREMENTS

- ServiceNow development/lab instance
- Administrator access (admin role)
- Excel / CSV spreadsheet editor
- Web browser
- Internet connection

7. TESTING REQUIREMENTS

The transform map configuration shall be verified using the following test cases:

- Load a valid Excel file containing records across all three branches (ECE, EEE, CSE).
- Run the transform map and confirm zero errors in the Import Set log.
- Verify target table (u_institution_details) population with auto-generated roll numbers and correct reference links.
- Test duplicate prevention by re-importing the same dataset with Coalesce configured.

8.EXPECTED OUTCOME

After completing the requirements, the system should provide:

- A functional Import Set staging table (u_institution_details_import).
- A fully mapped Transform Map (Institution Details Transform Map).
- Successfully transformed records in the u_institution_details table matching the source file.

9. CONCLUSION

The Requirement Analysis phase defines the staging structure, source-to-target field mappings, reference handling, and execution criteria for importing spreadsheet data into the Institution Details table using Transform Maps. These requirements provide the foundation for configuring the import set and executing the data load in the next phase.